Centennial Jing-Zhang AI Innovation Belt Urban Design

THE WARM RETURN

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Giving the heat of compute back to the city: Zhangbei green power → Haidian compute → urban waste heat, one energy chain as the organising motif of a nine-kilometre heritage park.

Overall design area	Key areas	Warm public nodes	Slow-mobility share
1,140.0 ha	**368.4 ha**	**10**	**86.8%**
announced	announced · 3 areas	recomputed	recomputed

Four self-imposed constraints

- Every number comes from 33 registered sources and 57 metrics: 47 recomputed, 10 left unknown.
- 12 assumptions are declared explicitly; the boundary is provisional geometry, not an official redline.
- All figures render offline from the submitted GeoJSON; no commercial basemap tiles.
- 25 self-checks, 0 blocking; uncertainty is written as a precondition, never as a conclusion.

Why the constraints come first

No verifiable, georeferenced version of this corridor's official redline or of the three key-area polygons has been published. A proposal that fills those gaps with attractive numbers leaves reviewers unable to tell which part to trust; a proposal that states the gaps on its cover lets them separate what can be relied on now from what must be redone once official data arrives. This booklet chooses the latter — every spatial recommendation is a conceptual proposal for professional teams to develop further.

Sheet index

Principal basis

- Haidian prequalification notice (scope and depth)
- Open-source call taskbook for global agents
- MOHURD urban design measures
- MOHURD control-plan measures
- MNR land and sea use classification guide

Evidence Chain and Package Map

The traceable path from public sources to submitted files | asserts no spatial boundary

Evidence Chain and Submission Package Map

Fig. 01 / Evidence

33 sources → 12 assumptions → 57 metrics → 9 layers and 5 figures | every step traces to a public source or is marked unknown

① Public sources

33

sources, each with publisher, URL, publication and access dates, licence and usage fence

- 26 official public sources, 25 at authority level A0
- each records not_usable_for to prevent over-reach
- foreign cases keep source conflicts and negative evidence, no averaging

→

② Declared assumptions

12

assumptions covering boundary, control-indicator and heat-data gaps

- boundary is provisional repository geometry, not an official redline
- five official controls are missing: FAR, height, density, green ratio, setback
- no public verifiable data on IT load or heat-network interfaces

→

③ Recomputed metrics

47/57

metrics known; the remainder are marked unknown rather than estimated

- 47 class-1 metrics recomputed from GeoJSON in EPSG:4548
- 8 class-2 metrics await official controls; no pseudo-precision
- 2 class-3 operational metrics; 10 unknown in total

→

④ Submitted artefacts

9+5

GeoJSON layers and 5 figures, all generated from the sources and assumptions above

- figures render directly from the submitted layers; no commercial basemap tiles
- bilingual proposal, report and figures correspond one to one
- 25 self-checks, 23 passed, 0 blocking

Submitted layers (feature count)

site_boundary	1	Provisional overall boundary
key_areas	3	Three provisional key areas
land_use	21	Land-use cells, non-overlapping and seamless
buildings	20	Building masses, incl. 3 retained
roads	12	Road and slow-mobility centrelines
green_space	6	Park and buffer green spaces
public_space	12	Station-city plazas and warm nodes
phasing	3	Near-, mid- and long-term phases
constraints	0	Deliberately empty: no verifiable public constraint lines

Three things this diagram is for

1. Traceability: every figure in the text carries a [source:], [metric:], [data:] or [standard:] marker and traces back to its file.
2. Recomputability: 47 class-1 metrics are recomputed from the submitted GeoJSON in EPSG:4548; the same layers yield the same values.
3. The right to refuse: 10 metrics stay unknown by choice. A number without official data is more dangerous than a blank.

This diagram describes the package's own structure and asserts no spatial boundary. Full contents are in sources.json, assumptions.json, metrics.json and self_check.json.

Why the evidence chain comes first

- The first risk in urban design is not form but citing data that does not exist.
- Sources, assumptions, metrics and artefacts form one traceable chain; every number returns to its file.
- Foreign precedents keep source conflicts and negative evidence rather than being averaged.

What is deliberately empty

- constraints.geojson holds zero features: with no verifiable public constraint lines, none are drawn.
- Five official controls are missing — FAR, height, density, green ratio, setback — and are left blank.
- Key areas reuse the repository's original rectangles and declare no precise area.

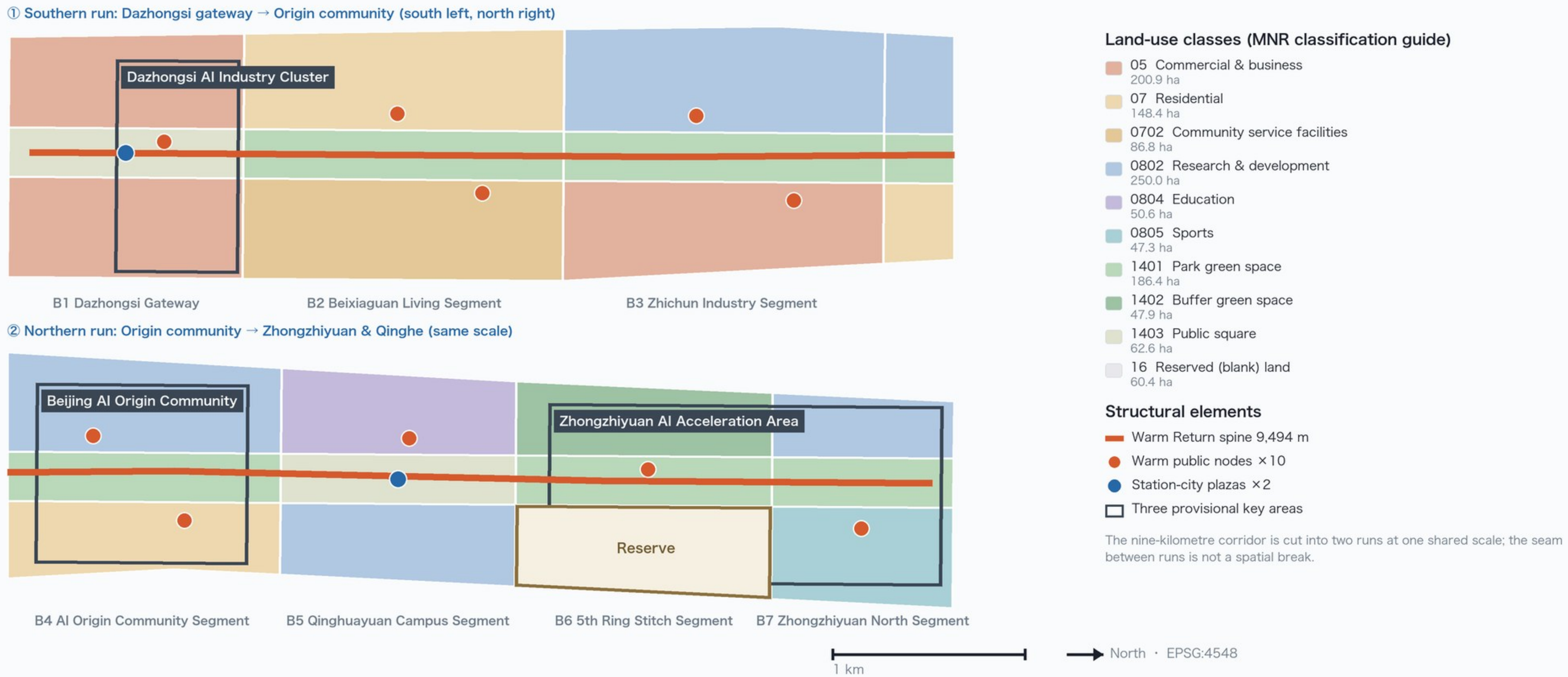
Three-Tier Scope Framework

Coordination study 43.6 km^2 / overall design 11.4 km^2 / key areas 368.4 ha

Overall Spatial Structure and Land-Use Layout

Fig. 02 / Land use

The Warm Return · one spine, three anchors, ten nodes, one reserved parcel | 21 seamless land-use cells, coverage 0.999993



Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released.

Three tiers are three kinds of question, not three scales

- The coordination scope answers the corridor's role in the AI map: completing the heat-to-city return leg.
- The overall design scope answers how the 1 - 2 km on either side of the heritage park renews and carries load.
- The key areas answer how plots, buildings, movement and public space are actually made.

One spine, ten nodes, three stitches, one blank

- The slow-mobility spine runs the full nine kilometres as a continuous walking experience.
- Ten warm public nodes give winter life somewhere to stay, recomputed at 5.1 ha.
- 60.4 ha of reserved land keeps a heat interface possible; no unfounded pipeline is drawn now.

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released.

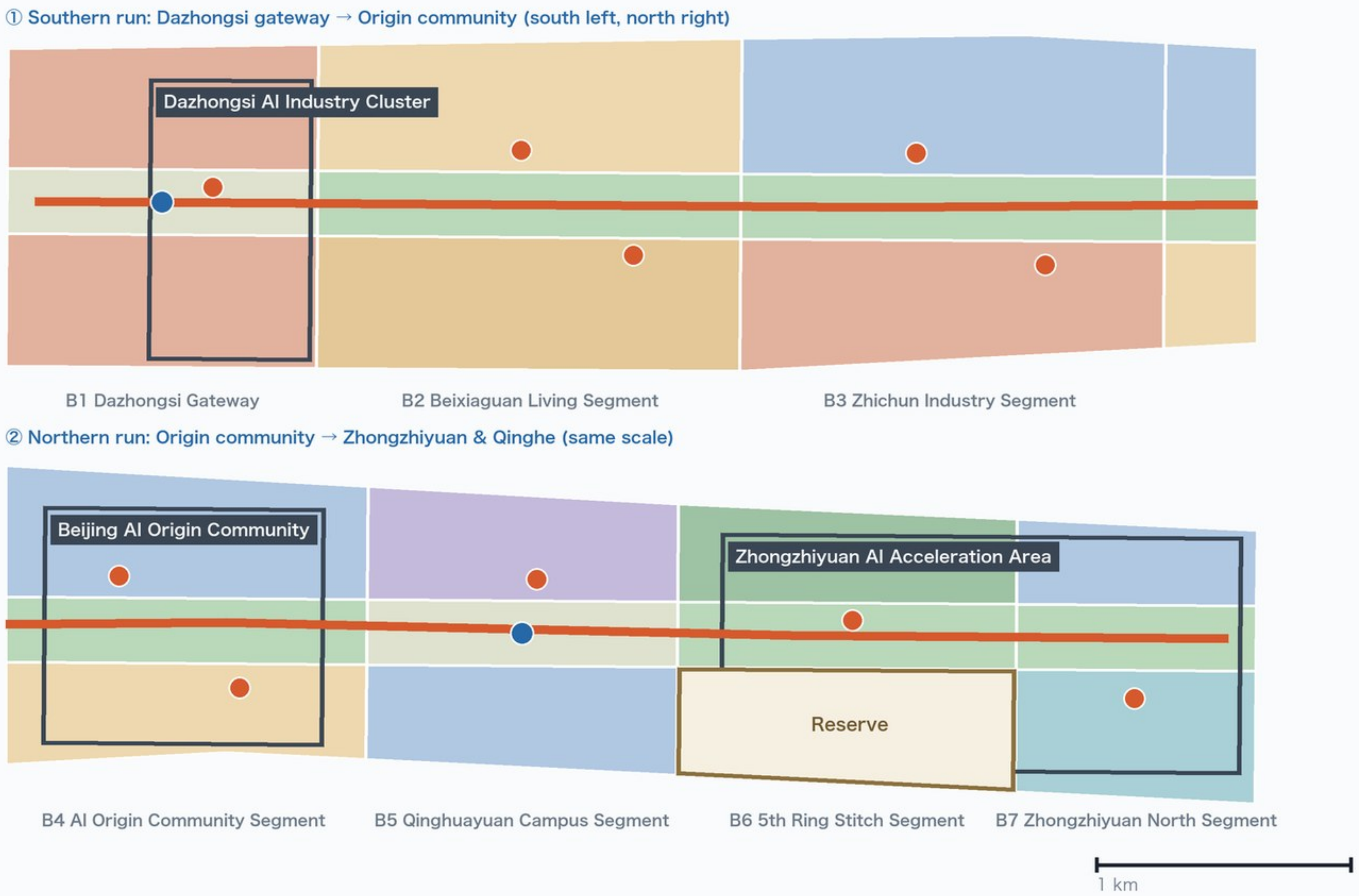
Overall Spatial Structure and Land Use

21 land-use cells cover the site seamlessly at 100.00% | the 9.7 km corridor is printed as two runs at one shared scale

Overall Spatial Structure and Land-Use Layout

Fig. 02 / Land use

The Warm Return · one spine, three anchors, ten nodes, one reserved parcel | 21 seamless land-use cells, coverage 0.999993



Land-use classes (MNR classification guide)

- 05 Commercial & business — 200.9 ha
- 07 Residential — 148.4 ha
- 0702 Community service facilities — 86.8 ha
- 0802 Research & development — 250.0 ha
- 0804 Education — 50.6 ha
- 0805 Sports — 47.3 ha
- 1401 Park green space — 186.4 ha
- 1402 Buffer green space — 47.9 ha
- 1403 Public square — 62.6 ha
- 16 Reserved (blank) land — 60.4 ha

Structural elements

- Warm Return spine 9,494 m
- Warm public nodes ×10
- Station-city plazas ×2
- Three provisional key areas

The nine-kilometre corridor is cut into two runs at one shared scale; the seam between runs is not a spatial break.

North · EPSG:4548

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released.

Three judgements behind the land-use structure

- 250.0 ha of R&D land concentrates mid-corridor, next to existing campuses and industry clusters.
- 186.4 ha of park land and 62.6 ha of squares form the warm-atrium spine.
- Housing plus community services total 235.2 ha, so the corridor is not a pure industrial belt.

What this sheet does not claim

- Cell boundaries are conceptual proposals, not plot redlines; not usable for approval or precise areas.
- No FAR or height is labelled: the official controls are missing, and filling them would be pseudo-precision.
- The seam between runs is a drafting cut, not a spatial break; both runs share one scale.

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released.

Key-Area Detailed Design Index

Three key areas announced at 368.4 ha, provisional geometry recomputes to 369.3 ha

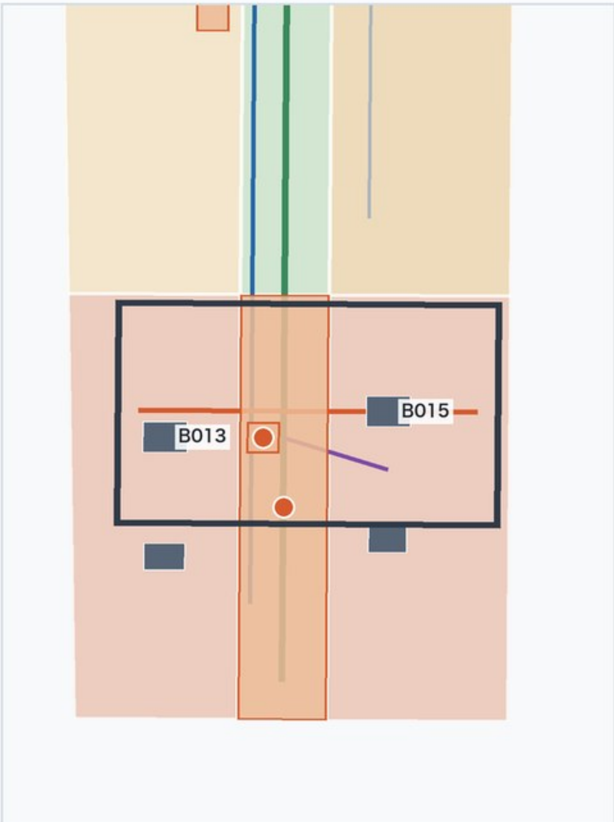
Three Key Areas: Detailed Design Index

Fig. 03 / Key areas

Provisional extents of the three announced key areas with this proposal's detailed design elements | all three panels share one scale, so areas and distances compare directly

1 | Dazhongsi AI Industry Cluster

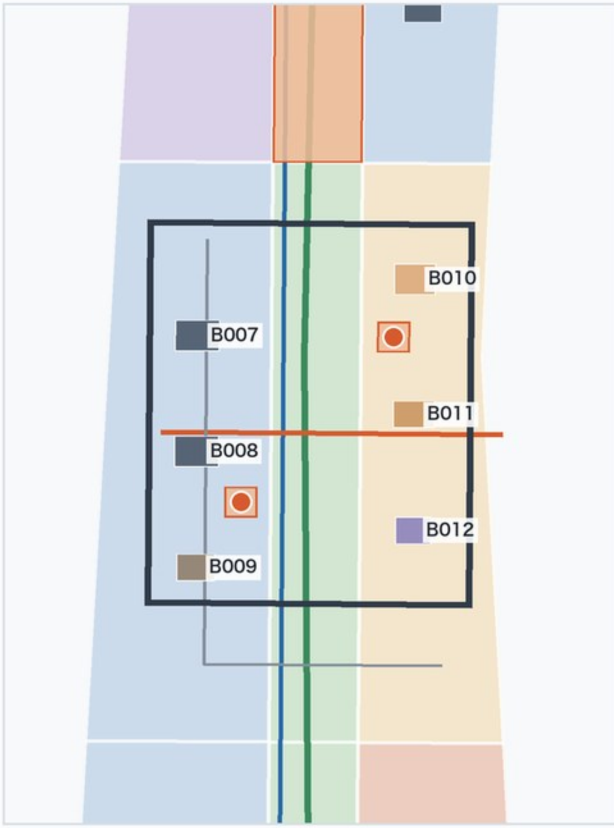
Announced ≈72.0 ha · recomputed 72.0 ha · rough inference



The city end: the rail gateway meets the commercial frontage; the station-city plaza hands metro flows straight to the Warm Return spine.

2 | Beijing AI Origin Community

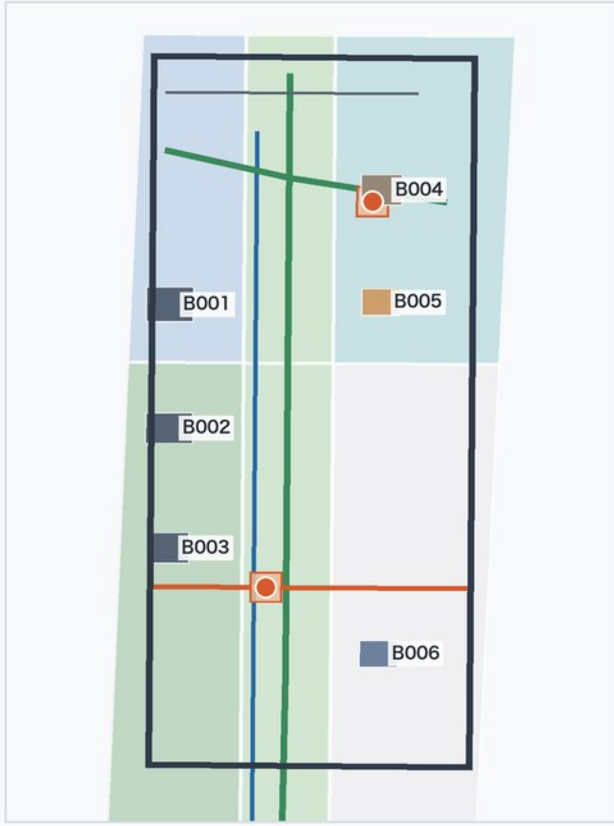
Announced ≈104.3 ha · recomputed 104.3 ha · rough inference



The human end: talent housing, incubation and campus-city links concentrate here; two warm lounges carry everyday dwelling and a micro-circulation branch carries arrival.

3 | Zhongzhiyuan AI Acceleration Area

Announced ≈192.1 ha · recomputed 192.9 ha · rough inference



Supply end: compute and heat sources sit here, so public space is closest to the technical plant. The reserved parcel adjoins it, holding room for heat exchange and municipal interfaces.

New or renewed mass (concept) · Retained buildings ×3 · Warm nodes and plazas · Greenway and spine · Pedestrian stitch · Transit connectors

500 m

North · EPSG:4548

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released. Building masses, heights and intensities are indicative only; control indicators await the statutory plan.

Each area carries one end of the energy chain

- Dazhongsi AI industry cluster: the city end, plugging compute services into existing retail and interchange.
- Beijing AI origin community: the human end, where community facilities and public space carry daily life.
- Zhongzhiyuan acceleration area: the supply end, with the densest R&D land and massing.

Announced and recomputed values printed side by side

- All three deviate by a few parts per thousand and serve only as an order-of-magnitude check.
- boundary_precision is provisional_rough and official_boundary is false.
- Once official polygons are released, every area and mass on this sheet must be recomputed.

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released.

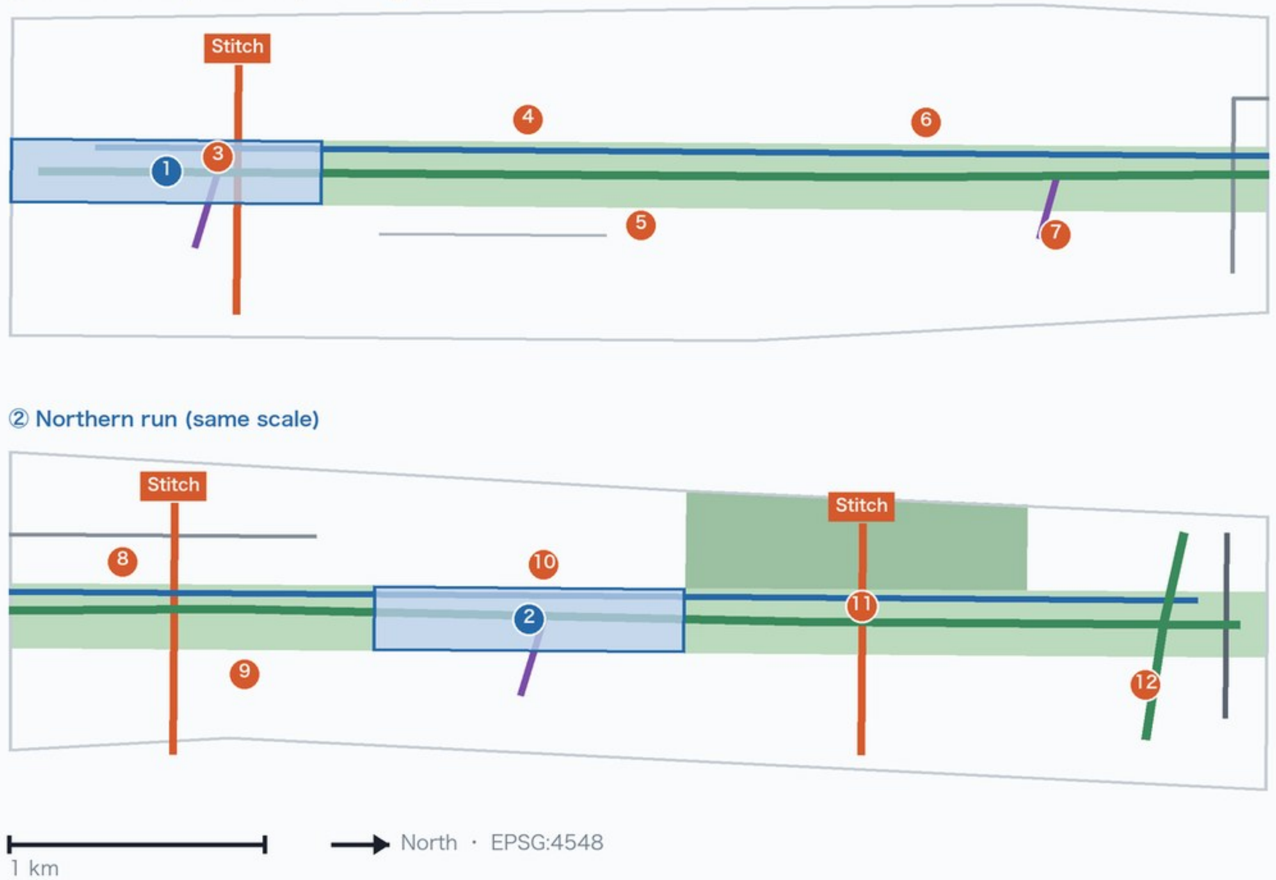
Slow Mobility and Blue-Green Public Space

Fig. 04 / Mobility & blue-green

One spine, three stitches, three transit connectors | slow mobility is 86.8% of network length; green and public space together 26.7%

① Southern run (south left, north right)

② Northern run (same scale)



Network hierarchy

- Greenway / slow-mobility spine — 10,319 m
- Continuous cycleway — 9,105 m
- Pedestrian stitch — 2,862 m
- Transit connector — 872 m
- Secondary road — 726 m
- Micro-circulation branch — 1,905 m
- Local access road — 888 m

Green and public space

- Park green space 1401 ×5 — 186.4 ha
- Buffer green space 1402 ×1 — 47.9 ha
- Station-city plazas ×2 — 62.6 ha
- Warm public nodes ×10 — 5.11 ha

Three things deliberately not drawn

This proposal does not propose rail alignment or station changes, does not estimate ridership, and does not draw heating pipe routes, diameters or connection capacities. All three depend on official data that is not public; drawing them would turn guesses into conclusions.

Public space node index

1. Dazhongsi Warm Gateway Plaza
2. Qinghuayuan Station Foyer
3. Dazhongsi Warm Room
4. Beixiaguan Warm Pavilion
5. Beixiaguan Service Point
6. Zhichun Heat Interface Hall
7. Zhichun Enterprise Lounge
8. Origin Open-Source Lounge
9. Origin Talent Lounge
10. Qinghuayuan Foyer Pavilion
11. 5th Ring Warm Gallery
12. Qinghe Warm Greenhouse

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released. Network lengths are recomputed from the submitted LineStrings in EPSG:4548 and exclude carriageway width or area.

Stitching before adding

- Three pedestrian stitches address where the corridor is severed, crossing rather than detouring.
- Three transit connectors bring stations within a single transfer of the heritage park.
- Green and public space together occupy 26.67% of the site, public space alone 6.14%.

Three lines deliberately not drawn

- No change to existing rail alignment: without approvals, drawing one would over-reach.
- No ridership or flow forecast: no verifiable public travel-survey data.
- No heat pipes: heat-network interface locations and capacities are not public.

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released.

Metric Recomputation and Evidence Chain

57 metrics: 47 recomputed, 8 awaiting official controls, 2 awaiting operational data

Metric Recomputation and Evidence Chain

Fig. 05 / Metrics

57 metrics in three classes: 47 recomputed from the submitted geometry, 8 awaiting official controls, 2 awaiting operational data | the latter 10 stay unknown, never estimated

The three metric classes

class 1 recomputable — 47

recomputed from GeoJSON in EPSG:4548

class 2 pending official control — 8

needs an approved control plan

class 3 operational performance — 2

needs IT-load and heat-network data

Confidence distribution (57 metrics)

high	10
medium	31
low	6
unknown	10

Low confidence clusters in building masses and key-area sizes: both derive from provisional geometry.

How anyone can independently recompute

① Read the nine GeoJSON layers in geometry/ (EPSG:4326, lon-lat order).

② Project to EPSG:4548 before computing area and length; never compute area in 4326.

③ Compare each result with the formula field in metrics.json and reproduce the same number.

④ A deviation above 0.5% counts as an error in this package; please raise it in PR review.

Main data files behind the metrics

geometry/land_use.geojson	12
brief/site-package/ranges/planning_limits.json	10
geometry/site_boundary.geojson	7
geometry/roads.geojson	7
geometry/public_space.geojson	6

class 1 recomputation sample (13 of 47)

site_area_sqm — 1,141.3 ha
polygon_area(site_boundary, EPSG:4548)
site_boundary.geojson

official_overall_design_area_sqm — 1,140.0 ha
official_announcement_value(overall_design_area)
planning_limits.json

site_area_deviation_from_official_ratio — 0.11%
abs(site_area_sqm - official_overall_design_area_sqm) / official_overall_d
site_boundary.geojson, planning_limits.json

land_use_coverage_ratio — 100.00%
area(intersection(union(land_use), site_boundary)) / site_area_sqm
land_use.geojson, site_boundary.geojson

land_use_area_0802_sqm — 250.0 ha
sum(polygon_area(land_use where land_use_code=0802))
land_use.geojson

green_ratio — 20.52%
green_space_area_sqm / site_area_sqm
green_space.geojson, site_boundary.geojson

public_space_ratio — 6.14%
public_space_area_sqm / site_area_sqm
public_space.geojson, site_boundary.geojson

warm_public_node_count — 10
count(public_space where public_space_type_zh='回暖公共节点')
public_space.geojson

road_centerline_length_m — 26.68 km
sum(line_length(road_centerline, EPSG:4548))
roads.geojson

slow_mobility_share_of_road_network_ratio — 86.80%
slow_mobility_length_m / road_centerline_length_m
roads.geojson

provisional_key_area_total_sqm — 369.3 ha
area(union(key_areas))
key_areas.geojson

official_key_detailed_design_area_sqm — 368.4 ha
official_announcement_value(key_detailed_design_area)
planning_limits.json

self_check_blocking_failures — 0
count(self_check.checks where result='fail' and severity='blocking')
self_check.json

class 2: 8 awaiting official control

building_density
No building survey or approved control plan

total_floor_area_sqm
Needs approved FAR and storey controls

floor_area_ratio
FAR is a statutory control, not derived by a proposal

building_height_control_m
Height limits need approvals (airspace, heritage)

pending_official_green_ratio_control
Official green-ratio control still pending

setback_control_m
Setbacks only follow the official road redline

road_area_sqm
No official redline or cross-sections; no road-area polygons

road_area_ratio
Both terms depend on the official road redline

class 3: why the two heat metrics must stay blank

waste_heat_recoverable_share_of_it_load_ratio
No public data on IT capacity, cooling type or return temperature

heat_supplied_households_equivalent
Heat load and network interfaces unknown; "N households" would be pseudo-precision

Waste-heat reuse is this proposal's central claim, so these two metrics are the ones a reader most wants — and precisely the ones that must not be invented. The recoverable share depends on installed IT capacity, cooling type, return temperature and the annual load curve; the household figure adds heat-network interfaces and heat demand on top. None of the four is verifiable in public sources.

What this proposal offers instead is a measurable set of interface conditions — heat-exchange room locations, utility-corridor reservations and end-user terminals — not an unverifiable recovery rate.

This diagram asserts no spatial boundary. All values are read from metrics.json and recomputed in EPSG:4548; everything must be recomputed once official boundaries and key-area polygons are released. 'unknown' is a deliberate choice, not an omission.

The recomputation protocol

- GeoJSON is stored in EPSG:4326; area and length are computed only after projection to EPSG:4548.
- The formula field in metrics.json states each algorithm so every value can be reproduced.
- A deviation above 0.5% counts as an error in this package; please raise it in PR review.

The two metrics a reader most wants must not be invented

- The recoverable share depends on installed IT capacity, cooling type, return temperature and the annual load curve.
- The household figure adds heat-network interfaces and heat demand; none of the four is publicly verifiable.
- What this proposal offers is a measurable set of interface conditions, not an unverifiable recovery rate.

Phasing, Risk and the Refusal List

Three phases cover the whole site | 12 declared assumptions | 25 self-checks, 0 blocking

Phasing and delivery logic

phase_1 — 317.1 ha — 27.8%

Near term: stitches and nodes first — low cost, visible quickly

phase_2 — 243.7 ha — 21.4%

Mid term: key-area detailed design and facility provision

phase_3 — 580.5 ha — 50.9%

Long term: reserved land and heat interfaces, once official data exists

Declared assumptions (12)

PROVISIONAL-SITE-BOUNDARY	major
PROVISIONAL-KEY-AREAS	major
MISSING-REGULATORY-CONTROLS	major
NO-OFFICIAL-ROAD-REDLINE	major
NO-CONSTRAINT-GEOMETRY	major
BUILDING-SAMPLE-NOT-SURVEY	major
RETAIN-RENOVATE-DEMOLISH-METHOD-ONLY	major
STATION-LOCATION-APPROXIMATE	minor
NO-COMPUTE-LOAD-DATA	blocking_for_quantified_claims
NO-DISTRICT-HEATING-NETWORK-DATA	blocking_for_quantified_claims
NO-OPERATIONAL-PERFORMANCE-DATA	minor
NO-GOVERNMENT-COMMITMENT	blocking_for_statutory_use

Each assumption states its impact and retirement condition, and is closed one by one once official data arrives.

The refusal list: ten kinds of conclusion this proposal declines to give

① The five statutory controls: FAR, density, height limit, green ratio, setback.

② Total floor area and per-plot floor area — no approved control plan or building survey.

③ Road-land area and share — no official redline or cross-sections.

④ Recoverable waste-heat share — no public data on IT capacity, cooling type or return temperature.

⑤ Households heated — heat load and network interfaces unknown; any "N households" is pseudo-precision.

⑥ Carbon reduction and energy-intensity figures — with no upstream data, no conclusion holds.

⑦ Ridership and section-flow forecasts — no verifiable public travel survey.

⑧ Adjustments to existing rail alignment — no approval basis.

⑨ Precise plot redlines and boundary coordinates — the submitted geometry is provisional and rough.

⑩ Investment estimates and land revenue — no public land price or cost basis.

All ten could be written beautifully and none could be verified. Reviewers can therefore tell which parts of this proposal can be trusted directly and which must be redone once official data exists — more useful than a proposal full of unverifiable numbers.

Pre-submission self-checks (25 items, 0 blocking)

Check	Status	Severity
SELF_CHECK_POLICY	pass	info
FORMAL_READINESS_THRESHOLD	pass	info
DATA_CONFIDENCE_ALIGNMENT	pass	info
METRIC_CRS_PROTOCOL	pass	info
BOUNDARY_TRUST	pass	major
KEY_AREAS_TRUST	pass	major
LAND_USE_TOPOLOGY	pass	blocking
LAND_USE_COVERAGE	pass	blocking
GEOMETRY_INSIDE_SITE	pass	major
METRIC_RECALCULATION	pass	blocking
UNKNOWN_METRIC_DISCLOSURE	pass	major
CONSTRAINTS_DATA_GAP	unknown	major
LOCKED_LAYER_RESPECT	pass	blocking
BUILDING_SAMPLE_DISCLOSURE	pass	major
PHASING_COVERAGE	pass	minor
UNQUANTIFIED_HEAT_CLAIMS	pass	blocking
NO_GOVERNMENT_COMMITMENT_CLAIM	pass	blocking
PERSONAL_DATA_MINIMIZATION	pass	major
HUMAN_FALLBACK_DESIGN	pass	major
VISUAL_STATIC	pass	blocking
VISUAL_METRIC_CONSISTENCY	pass	major
BILINGUAL_PAIRING	pass	blocking
COPYRIGHT_CLEARANCE	pass	blocking
PROFESSIONAL_EVIDENCE	pass	major
OFFICIAL_DATA_RECOMPUTE_PLEDGE	unknown	major

All nine blocking-severity checks concern provisional geometry and missing official controls; this package writes them into the text, metrics and drawings rather than making them disappear.

Copyright, licence and ongoing maintenance

- All text, layers, figures and PDFs were generated by an AI agent (Ducc / Claude Opus 5) within a single session, using no non-public material and no commercial basemap tiles.
- Licensed COMMUNITY-DISPLAY-ONLY: usable for community display and review discussion; not a planning deliverable and not an approval basis.
- Once official boundaries or control conditions are released, this package commits to re-running geometry, metrics, figures and all four validators, recording the differences in the changelog.
- Every figure is rendered by script straight from metrics.json / sources.json / assumptions.json / self_check.json, so drawings and data files cannot diverge.

33	12	25	47 / 57	10
Public sources	Assumptions	Self-check items	Recomputable metrics	Deliberate unknowns